

16/05/19 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam**Max. Marks: 100****May-June 2019****Duration: 3 Hrs**

Class: S Y BTech

Semester: IV

Name of the Course: Applied Mathematics-IV

Branch: COMP/IT

Course Code: UCEC401 / UIRC401

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.	Questions										Marks																															
Q.1	A	Find the coefficient of correlation between x and y for the following data										5																														
	<table><tr><td>x</td><td>62</td><td>64</td><td>65</td><td>69</td><td>70</td><td>71</td><td>72</td><td>74</td></tr><tr><td>y</td><td>126</td><td>125</td><td>139</td><td>145</td><td>165</td><td>152</td><td>180</td><td>208</td></tr></table>										x		62	64	65	69	70	71	72	74	y	126	125	139	145	165	152	180	208													
	x	62	64	65	69	70	71	72	74																																	
	y	126	125	139	145	165	152	180	208																																	
	B	Solve any Two of the following.										16																														
(i)	Ten competitors in a beauty contest are ranked by three judges in the following order																																									
<table><tr><td>Judge I</td><td>1</td><td>6</td><td>5</td><td>10</td><td>3</td><td>2</td><td>4</td><td>9</td><td>7</td><td>8</td></tr><tr><td>JudgeII</td><td>3</td><td>5</td><td>8</td><td>4</td><td>7</td><td>10</td><td>2</td><td>1</td><td>6</td><td>9</td></tr><tr><td>JudgeIII</td><td>6</td><td>4</td><td>9</td><td>8</td><td>1</td><td>2</td><td>3</td><td>10</td><td>5</td><td>7</td></tr></table>										Judge I	1		6	5	10	3	2	4	9	7	8	JudgeII	3	5	8	4	7	10	2	1	6	9	JudgeIII	6	4	9	8	1	2	3	10	5
Judge I	1	6	5	10	3	2	4	9	7	8																																
JudgeII	3	5	8	4	7	10	2	1	6	9																																
JudgeIII	6	4	9	8	1	2	3	10	5	7																																
Use rank correlation coefficient to determine which pair of judges has the nearest approach to common tastes in beauty.																																										
(ii)	The two regression lines are $4x - 5y + 33 = 0$; $20x - 9y = 107$ and variance of x is 25. Find i) mean of x & y ii) coefficient of correlation iii) variance of y																																									
(iii)	Fit the second degree curve to the following data																																									
<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>y</td><td>2</td><td>6</td><td>7</td><td>8</td><td>10</td><td>11</td><td>11</td><td>10</td><td>9</td></tr></table>										x	1	2	3	4	5	6	7	8	9	y	2	6	7	8	10	11	11	10	9													
x	1	2	3	4	5	6	7	8	9																																	
y	2	6	7	8	10	11	11	10	9																																	
Q.2	A	Two unbiased dice are thrown. Let X_1 and X_2 denote the scores obtained and Y denote the maximum of them i.e. $Y = \max.(X_1, X_2)$. Find the probability distribution of Y.										5																														

	B	Solve any Two of the following.	16
	(i)	A company owns 400 laptops. Each laptop has an 8% probability of not working. You randomly select 20 laptops for your sales people. (a) What is the likelihood that 5 will be broken? (b) What is the likelihood that they will all work? (c) What is the likelihood that they will all be broken?	
	(ii)	In a normal distribution 7% of items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution?	
	(iii)	The number of traffic accidents that occurs on a particular stretch of road during a month follows a Poisson distribution with a mean of 9.4. Find the probability that less than two accidents will occur on this stretch of road during a randomly selected month.	
Q.3	A	A random sample of 50 items gives the mean 6.2 and S.D. 10.24. Can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?	5
	B	Solve any Two of the following.	
	(i)	According to a theory the proportion of a commodity in the four classes A, B, C, D should be 9 : 4 : 2: 1. In a Survey of 1600 items of this commodity the numbers in the four classes were 882, 432, 168 and 118. Does the survey support the theory?	
	(ii)	A random sample of size 16 has mean 53 and sum of square deviations taken from mean is 150. Can it be regarded as drawn from a population with mean 56?	16
	(iii)	The mean life of a sample of 100 electric light bulbs was found to be 1456 hours with S.D. 400. A second sample of 225 bulbs chosen from a different batch showed a mean life of 1400 hours with standard deviation of 144 hours. Assuming that the two populations have same standard deviation find, if there any significant difference between the mean of two batches?	
Q.4		Solve any Two of the following.	16
	(i)	Consider a self-service store with one cashier. Assume Poisson arrivals and exponential service times .Suppose that an average nine customers arrive every 5 min. and that the cashier can serve 10 in 5 min. Find: (a) average number of customers in the system (b) probability of having more than 10 customers in the system, and (c) probability that a customer has to queue for more than 2 min. If the service can be speed up to 12 in 5 min. by using a different cash register, what will be the effect of this on the quantities (a), (b) and (c)	

	(ii)	At a railway station, only one train is handled at a time the railway yard is sufficient only for two trains to wait while the other is given a signal to leave the station. Trains arrive at the station at an average rate of 6 per hour and the railway station can handle them on an average 12 per hour. Assuming Poisson arrivals and exponential service distribution, find the steady state probabilities for the various number of trains in the system. Also find the average waiting time of a new train arriving at the yard.	
	(iii)	A maintenance service facility has Poisson distribution arrival rates, negative exponential service times, and operates on a first come, first served queue discipline. Breakdowns occur on an average of three per day. The maintenance crew can service, on an average, six machines per day, Find the: (a) Utilization factor of the service facility (b) Mean waiting time in the system (c) Mean number machines in the system (d) Mean waiting time of machines in the queue	
Q.5	A	Maximize $Z = 4x_1 + 3x_2 + 6x_3$ Subject to: $2x_1 + 3x_2 + 2x_3 \leq 440$ $4x_1 + 3x_3 \leq 470$ $2x_1 + 5x_2 \leq 430$ $x_1, x_2, x_3 \geq 0$	5
	B	Solve any Two of the following.	16
	(i)	Solve the following L.P.P. by Big-M Method Maximize $Z = 3x_1 - x_2$ Subject to: $2x_1 + x_2 \geq 2$ $x_1 + 3x_2 \leq 3$ $x_2 \leq 4$ $x_1, x_2 \geq 0$	
	(ii)	Using the method of Lagrange's multipliers, solve the following NLPP Optimize $z = 2x_1^2 + 2x_2^2 + 2x_3^2 - 24x_1 - 8x_2 - 12x_3 + 260$ Subject to $x_1 + x_2 + x_3 = 11, x_1, x_2, x_3 \geq 0$	
	(iii)	Use the Kuhn-Tucker conditions to solve the following NLPP Maximize $z = 16x_1 + 6x_2 - 2x_1^2 - x_2^2 - 17$ Subject to $2x_1 + x_2 \leq 8, x_1, x_2 \geq 0$	

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End Semester Exam
April - May 2019

Max. Marks: 100

Class: S.Y.BTech

Name of the Course: Microprocessor

Branch: Computer Engineering

Duration: 3 Hrs.

Semester: IV

Course Code: UCEC402

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Draw and explain read and write timing diagram when 8086 processor works in minimum mode.	10M
Q 1 (b)	Why we need to connect 8284 along with 8086? Explain power on reset circuit of 8284.	10M
Q 2	Design a 8086 based system consisting of the following: a) 8086 microprocessor working at 6 MHz b) 8087 math co-processor for performing complicated math. c) 128 KB EPROM using 32 KB d) 32 KB RAM using 8 KB devices. e) 2, 16 bit input and 2, 16 bit output ports in handshaking mode. f) Draw memory map and I/O map and complete design.	20M
Q3 (a)	Write a program in assembly language to check whether the string is palindrome or not. If the string is palindrome display the appropriate message.	10M
Q3 (b)	Compare and contrast closely coupled and loosely coupled system.	10M
Q4 (a)	What is an Interrupt? Explain different types of interrupts supported by 8086.	10M
	OR	
Q4 (a)	List and describe the steps an 8086 will take when it responds to an interrupt.	10M
Q.4 (b)	Explain various data transfer modes of 8237 DMA controller.	10M
	OR	
Q.4 (b)	Draw functional block diagram of 8255 and explain its working.	10M
Q5 (a)	Draw the block diagram of 80386DX processor and explain each block in brief.	10M

	OR	
Q5 (a)	Draw and explain EFLAG register of x86 processors.	10M
Q5 (b)	Explain integer instruction pipeline stages of Pentium processor. List the steps in instruction issue algorithm.	10M
	OR	
Q5 (b)	Explain dynamic branch prediction logic of Pentium processor in brief.	10M

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End Semester Exam
May-June 2019

Max. Marks: 100

Class: SYBTECH

Name of the Course: Relational Database Management Systems

:Computer

Course Code: UCEC404

Duration: 3 hrs

Semester: IV

Branch

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	List the disadvantage of the file processing system. Explain how database system can overcome this.	10 M
Q 1 (b)	Draw EER model for hospital management system. Assume required data.	10M
Q2 (a)	Consider the database: A library has the following relations: Library (Code, Name, No_of_Books) Person (id , name ,age) IS_Member(Code ,id ,Date _of_joining) BOOKS(Accession_No, Title, Author,Price) Borrowed_By (Accession_No,id ,Date _of_borrow) <i>Write SQL queries for the following.</i> a) List all the books borrowed before 2nd March 2019 and details of borrowers. b) List the details of books and persons who has borrowed the book costing above RS 2,000. c) Give the details of person who have not borrowed any book. d) List the person who have borrowed at least all the books borrowed by person with id 201.	10 M
Q2 (b)	Explain the rules to map ER schema to relation with example.	10 M

Q3 (a)	What is a View ? What are the benefits and limitations of View? OR What is transaction? Explain its properties.	10 M
Q3(b)	What is the difference between multivalued dependency and functional dependency. OR Explain referential integrity and its relation with foreign key.	10 M
Q4 (a)	Explain log based recovery technique in detail. OR Describe deadlock management techniques in databases.	10 M
Q4(b)	Explain timestamp ordering protocol with example.	10 M
Q5	Write short notes on (any 2) i) Conflict serializability ii) Relational Algebra iii) Triggers iv) Database administrator	20 M

28.05.19(19)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
May-June 2019

Max. Marks: 100**Class: SY****Name of the Course: Digital Communication and Networks****Course Code: UCEC405****Duration: 3 Hrs****Semester: IV****Branch: COMP****Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Compare and contrast HDLC with PPP in any five ways.	10
Q 1 (b)	List two ways in which the OSI reference model and the TCP/IP reference model are the same. Now list three ways in which they differ.	10
Q 2 (a)	Discuss the types, causes and effects of the various forms of noise which may be created within a receiver.	10
Q 2 (b)	Differentiate i) connectionless communication and connection-oriented communication and ii) Reliable and Unreliable services	10
Q 3 (a)	Exemplify the following two mechanisms for modulating digital data into an analog signal: i) ASK and ii) FSK.	10
Q 3 (b)	Differentiate Physical Network Address and Logical Network Address in five ways.	10
Q 4 (a)	Given the dataword 1010011110 and the divisor 10111 i) Show the generation of the codeword at the sender site (using binary division) ii) Show the checking of the codeword at the receiver site (assume no error). OR Using 5-bit sequence numbers, Find and justify the maximum size of the send and receive windows for each of the following protocols? i) Go-Back-N ARQ ii) Selective-Repeat ARQ	10

Q 4(b) Explain slotted aloha in detail.

Specify how slotted aloha overcomes problems of pure aloha.

OR**10****Q 4(b) Describe Adaptive Delta Modulation in detail.**

Q 5 (a)	Show the bit stuffing at the sender and bit removal at the receiver for the following data from upper layer. 000111111100111110100011111111111000011111 OR	10
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P.T.O.

	Q 5(a) Write a note on different types of spread spectrum.	10
Q 5 (b)	Discuss any two IEEE Ethernet technologies. OR Differentiate CSMA/CD and CSMA/CA Random Access Protocols in any five ways.	10